

REVIEW

Cardiac Implantable Electronic Devices Remote Programming

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ABSTRACT: Remote monitoring has revolutionized the follow-up of cardiac implantable electronic device recipients, leading to both organizational benefits and improved patient survival. To fully realize the potential of a remote cardiac implantable electronic device management strategy, a 2-way communication system enabling remote programming capabilities is essential. In recent years, several initiatives have emerged to develop and evaluate indirect remote programming solutions based on the remote control of a device programmer. This hybrid approach, which mitigates cybersecurity risks, has been successfully applied in various clinical settings, from magnetic resonance imaging procedures to medically underserved regions, and has consistently demonstrated high safety and patient satisfaction. Indirect remote programming reduces the need for both patient and healthcare professional travel, thereby lowering associated costs and the carbon footprint, while shortening delays and enhancing patient safety. It provides a clear preview of the enormous potential of direct remote programming, particularly regarding equity in access to care and patient outcomes. Direct remote programming of cardiac implantable electronic devices is no longer technical a barrier and is already available for implantable cardiac monitors. In the near future, extending direct programming to therapeutic cardiac implantable electronic devices is expected to further remove geographic and logistical barriers, reduce time to intervention, and improve patient safety. This promising evolution will, however, require not only addressing cybersecurity challenges but also embracing a paradigm shift within the cardiology community.

Key Words: computer security ■ defibrillators, implantable ■ pacemaker, artificial ■ remote consultation ■ remote patient monitoring ■ telemedicine

Remote monitoring (RM) of cardiac implantable electronic devices (CIEDs) is now widely regarded as a standard of care endorsed by multiple international guidelines and expert consensus documents.¹ By enabling early notification of clinically actionable events, RM offers substantial clinical benefits, including reduced hospitalizations and improved survival rates.^{2,3} However, in some instances, the benefits of RM are hindered by the difficulty or lack of timely device reprogramming. Remote programming (RP), that is, the ability to adjust device settings remotely without requiring a patient to visit the clinic, has long been the aspiration of some specialists dealing with CIED RM. In recent years, some manufacturers and even cardiologists have taken steps to make this a reality.^{4–11} Two forms of RP have evolved: indirect RP (IRP), which has been preferred for therapeutic devices (due to cybersecurity concerns), and direct RP (DRP), almost

exclusively used for implantable cardiac monitors.^{12,13} IRP of CIEDs refers to the remote control by a CIED specialist of a programmer that is connected locally to the patient's CIED in a clinical facility without specialist CIED support. This hybrid approach, which has the advantage of mitigating security risks, has been explored in various clinical scenarios and was consistently associated with high levels of safety and patient satisfaction. Benefits have also been shown for patients living in medically underserved areas, suggesting that IRP holds the potential to improve equity in access to care and reduce the carbon footprint. DRP is expected to further amplify these organizational advantages and play a key role in emergency situations due to improved response times. However, this potential can only be realized if the substantial challenges associated with cybersecurity, operational safety, and clinical workflow integration are systematically addressed.

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Nonstandard Abbreviations and Acronyms	
CIED	cardiac implantable electronic device
DRP	direct remote programming
ICD	Implantable Cardioverter Defibrillator
ICM	implantable cardiac monitor
IRP	indirect remote programming
MRI	magnetic resonance imaging
RM	remote monitoring
RP	remote programming

This review explores the rationale, current applications, associated advantages, and limitations of indirect and direct remote CIED programming, with a focus on security, feasibility, and clinical benefit.

NEED FOR CIED RP

One does not realize the necessity for CIED RP until one gains familiarity with CIED RM. In patients with CIEDs, RM has been recommended as the standard of care since the 2015 HRS Expert Consensus Statement, a recommendation that has been reinforced in the 2023 version.^{1,14} These recommendations are supported by several randomized trials as well as large registries, which have consistently demonstrated major organizational benefits and improved clinical outcomes when compared with scheduled in-office follow-up. At least 4 randomized trials have highlighted the utility of the latter follow-up strategy, since 71% to 98.5% of the scheduled visits were found to be noncontributory (ie, no change in patient management or device reprogramming).^{15–18} Nevertheless, in-person CIED testing and reprogramming are still required in patients with automatic and comprehensive lead monitoring up to every 24 months.¹ This time interval has been validated in the above-mentioned trials and allows for manual verification of the lead electrical parameters and resetting of the alert counter between 2 programmer interrogations. More frequent in-person visits may be necessary for patients with inconsistent connectivity or devices lacking automatic functions for capture/sensing testing.¹⁹ On top of these residual scheduled interrogations, unplanned interrogations are sometimes required, either prompted by a clinical issue (symptoms, surgery, magnetic resonance imaging [MRI]) or by an RM alert. MRI scanning is a typical situation in which RP may help address an unmet need. At present, nearly half of hospitals do not perform MRI scans in patients with CIEDs, with an estimated 10-fold national shortfall in the United Kingdom.^{20,21} The most commonly cited reason for denying MRI to CIED recipients is the lack of cardiology support (64%). Implementing RP may improve access to MRI, potentially resulting in important

changes in diagnosis, prognosis, and clinical management for this population. Although RM is not affected by physical barriers, distance can be an issue when it comes to reprogramming, particularly in some emergency situations, as presented in Figures 1 and 2.

INDIRECT REMOTE PROGRAMMING

Pacemakers and Implantable Cardioverter Defibrillators (ICDs) that provide life-sustaining therapy pose unique clinical and cybersecurity challenges compared with purely diagnostic cardiac devices. As a result, commercially available systems have traditionally offered only limited DRP functionality. Instead, a pragmatic approach of IRP has emerged, which operates by remotely controlling the CIED programmer rather than the implanted device itself. This distinction is crucial, as it significantly reduces the associated cybersecurity risks. Operationally, a bedside operator, typically a nonspecialist such as a nurse or technician, interfaces directly with both the patient and the programmer. Concurrently, a remote electrophysiologist or industry representative provides expert supervision, often maintaining real-time communication via telephone or video conferencing. The Table summarizes the results of the currently published experiences with IRP.

TECHNOLOGIES

Proprietary Solutions

The first experience with IRP was acquired using the Remote Control application (Medtronic, Minneapolis, MN) installed on a 2090 programmer with a modem card for internet connectivity.⁴ After a session key exchange between the 2 users, the programmer could be remotely controlled by a laptop using a third-party website (BeyondTrust Corporation, Atlanta, GA—formerly Bomgar), which ensures encrypted communication. The local staff initiates device connectivity through the standard programmer interface. The remote device specialist then performs the usual device functions and assesses the data from any location. This solution received US Food and Drug Administration approval in 2015 but was officially decommissioned in 2025 (and therefore is no longer used).

Tong et al⁸ tested a research tool that allowed remote testing and programming of Abbott devices. A 5G remote support terminal (China Telecom Corporation Limited Shanghai Branch, Shanghai) was connected to the conventional programmer (Merlin Patient Care System Model 3650), enabling IRP on a tablet with a 5G cloud follow-up application. The system used 2-step login and end-to-end cryptographic communication protocols, as well as a private cloud, to ensure cybersecurity.

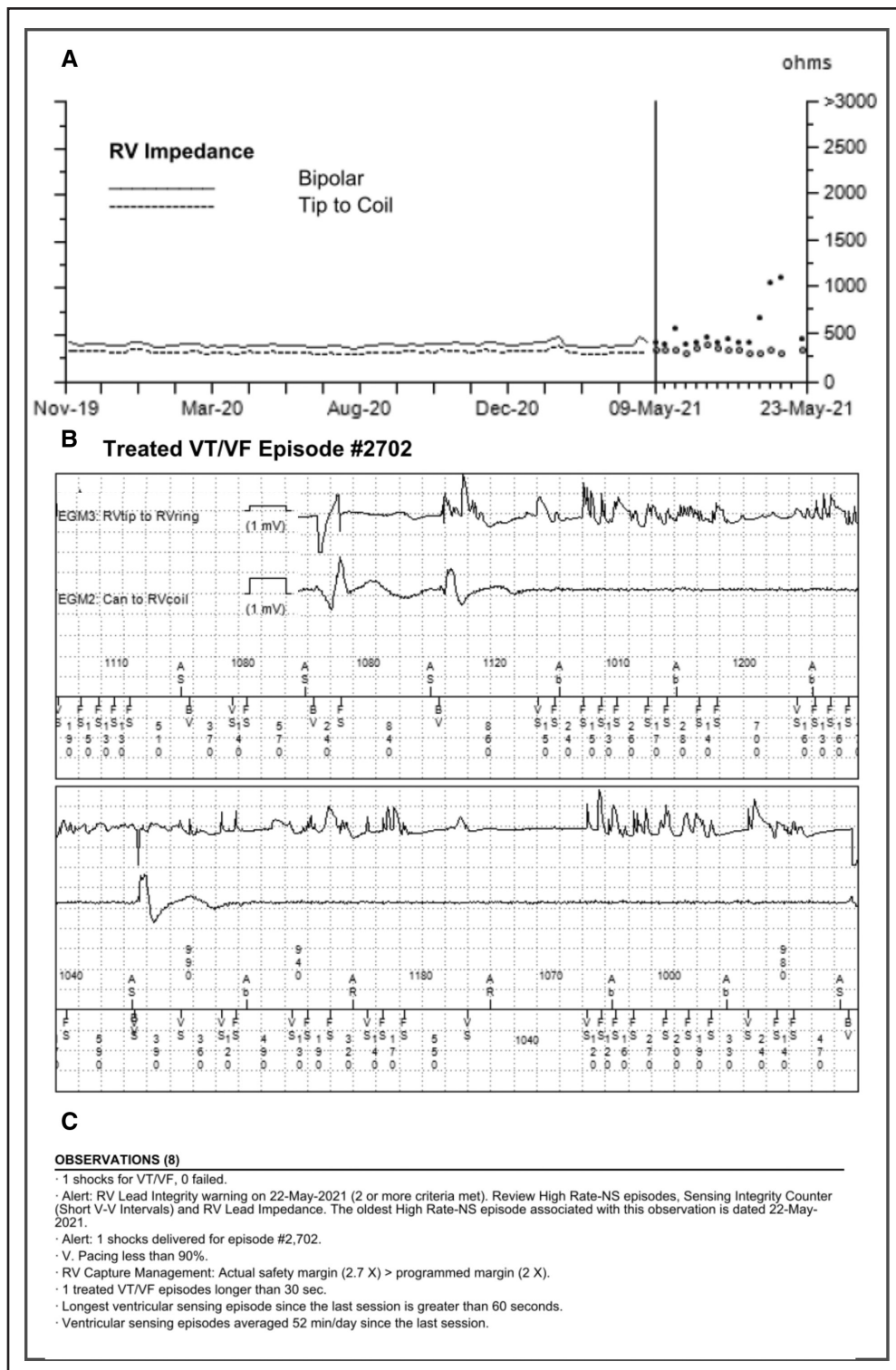


Figure 1. An illustrative case of remote lead failure diagnosis with a near fatal intervention delay.

A 77 year-old woman with dilated cardiomyopathy and complete left bundle branch block received a Medtronic Amplia CRT-D in 2017 and underwent atrioventricular (AV) nodal ablation for atrial fibrillation in 2020. On May 5, 2021, while the patient was vacationing abroad, a CareLink remote alert indicated right ventricular (RV) lead failure (Medtronic 6935 Sprint Quattro). Findings included an abrupt impedance rise (380→990 Ω), ventricular oversensing typical of conductor fracture, multiple short RR intervals (<130 ms), RV loss of capture, and noise detection at 1.1 mV—resulting in asystolic events in this pacing-dependent patient. Despite repeated remote alerts and explicit reprogramming instructions (switching sensing from bipolar to tip-to-coil), no adjustments were made during local in-office evaluations on May 10 and 17. Oversensing episodes and asystole increased, culminating in an implantable cardioverter defibrillator (ICD) shock on May 25. At that point, the patient was unable to stand anymore because of incessant episodes of syncope. The nightmare finally ended on May 27 when a Medtronic representative guided remotely by our remote monitoring (RM) center via WhatsApp implemented the recommended reprogramming. The patient was then transferred to Paris for RV lead replacement. (Continued)

Two proprietary systems currently offer IRP capabilities through commercially available programmers. The Biotronik Renamic Neo with the Live Support function has obtained FDA market authorization and CE marking under the EU Medical Device Regulation.²² This functionality enables clinicians to obtain real-time technical assistance from Biotronik specialists during in-clinic follow-up. The Abbott Merlin 2 Remote Support feature was recently granted FDA approval and is currently in limited market release in the United States of America. This system allows an authorized Abbott representative to provide direct programmer support to a local clinician through the Merlin Live Connect web application. In both systems, secure communication is initiated following the exchange of a randomly generated one-time password.

Multivendor Solution

We recently developed and validated a simple, yet highly effective, multivendor solution for the remote interrogation and reprogramming of CIEDs. In practice, this IRP system is operated by a cardiologist working from a workstation that remotely controls a local computer connected to a conventional programmer.^{10,11}

A first version of the system was compatible with the Abbott, Biotronik, and MicroPort programmers from which the screen was streamed to a local computer using a video capture device (Epiphan, Ottawa, Canada, VGA/HDMI to USB). The cursor of the programmer screen was controlled with a microcontroller (Arduino, Monza, Italy), which was connected between the local computer and the programmer by USB cables. Third-party video-conferencing software (TeamViewer, Stuttgart, Germany) was used to enable communication between the computer used for the in-person visit and the remote-control workstation via the internet.

Rather than a laptop, the advanced version of our system uses a minicomputer with no screen housing a customized Raspberry Pi. This microcomputer can be connected to the CIED programmer of the 5 main manufacturers (the Boston Scientific programmer needs to be unlocked by the manufacturer) via either VGA or HDMI for screen capturing, while a USB cable is used for cursor control. A secure virtual private network connection (Tailscale) facilitates communication between the minicomputer and the remote workstation via the internet, whether over Wi-Fi or 4G. This virtual private

network provides end-to-end encryption, safeguarding all transmitted data from potential interception or unauthorized access. Whitelisting maintains an approved list of users or IP addresses, limiting access and ensuring that only authorized personnel can connect to the system (Figure 3).

EXPERIENCE WITH IRP

IRP has been used across a variety of clinical scenarios, among which are pre- and post-MRI scan programming, scheduled follow-up, emergency interrogations, either clinically or alert-driven, and technical support during device implantation procedures in the electrophysiology (EP) laboratory (Table).

Traditionally, MRI procedures in patients with CIEDs require the on-site presence of a healthcare professional trained in device programming to manage the device both before and after imaging based on physician-prescribed MRI-safe programming mode.²³ Given the inherent need for device reprogramming surrounding MRI studies, the MRI environment represented an ideal opportunity for the initial experience with RP of CIEDs. Using the Medtronic Remote Control system, Kloosterman et al⁴ successfully completed 100 RP sessions across 50 MRI procedures, with both pre- and postscan reprogramming performed without complication. Using a similar system, Siddamsetti et al⁶ subsequently reported comparable results in a single-center observational study involving 209 consecutive patients undergoing MRI (418 IRP sessions). Neither study involved specialized staff performing on-site programming. This strategy resulted in significant time savings by eliminating the need for on-site representatives or cardiologists, without compromising procedural safety or efficiency.

Tong et al⁸ assessed Abbott device IRP in the context of in-hospital follow-up in a region of China where RM was not available and compliance with conventional in-person evaluation was limited (60%). A total of 325 devices (91% pacemakers) from 13 hospitals were successfully interrogated, tested, and programmed remotely by an electrophysiologist at the Hospital of Chengdu. No complication occurred, and 90% of the patients chose cloud follow-up for further device management.

To demonstrate the benefits of IRP for patients with scheduled follow-up appointments who live in a medically underserved area, we held 34 IRP sessions in a

Figure 1 Continued. A simple remote programming (RP) of ventricular sensing polarity could have prevented 3 weeks of life-threatening pacing inhibition during which the device recorded 685 episodes of nonsustained ventricular tachycardia (NSVT) and 1659 episodes of ventricular sensing episodes corresponding to as many cases of asystole. The illustration includes samples of the remote transmission from the May 23, 2021 (18 days from initial diagnosis). **A**, Impedance trends of the lead showing abrupt increases in impedance limited to the bipolar dipole, in favor of a fracture of the ring conductor. **B**, EGM from the inappropriate shock episode. The upper channel corresponds to the RV Tip to Ring channel with electrograms typical of a conductor fracture. The lower channel corresponds to the Can to Coil channel and illustrates the true underlying rhythm of the patient with a pause exceeding 6s. Note the RV loss of capture on the first biventricular (BV) paced QRS complex. **C**, Observations list from the transmission. The ventricular oversensing caused by the ring conductor fracture cumulates 52 min/d which explains the drop in ventricular pacing (<90%). PAC indicates premature atrial complex; and RVP-MT, right ventricular pacing at maximal tracking rate; VF, ventricular fibrillation; and VT, ventricular tachycardia.



A VISIONIST CRT-P transmission received on February 25, 2026 contained 2 alerts for nonsustained ventricular arrhythmia and ventricular arrhythmia episodes, occurring on 21 and 16 February, respectively, in a 25-year-old patient implanted with an epicardial device for congenital atrioventricular block associated with congenitally corrected transposition of the great arteries. The recorded EGM showed nonsustained ventricular fibrillation episodes initiated by a long cycle-short cycle ventricular pacing sequence triggered by transient atrial oversensing caused by electromagnetic interference. The patient was unsuccessfully contacted on 3 occasions. Her mother was eventually reached and informed that the patient required urgent pacemaker reprogramming. Despite our explanations and warnings, both the patient and her mother considered it inconvenient to travel to our center, located 105 km away (approximately a 1 h 30 min drive). The patient was finally evaluated 24 hours later. No external source of interference could be identified during the patient interview. The device was temporarily programmed in DDIR mode pending further investigation of the patient's environment. Direct remote programming could have secured the patient more rapidly, while avoiding travel and the loss of a working day for both the patient and her mother. PAC indicates premature atrial complex; and RVP-MT, right ventricular pacing at maximal tracking rate.

distance and cost (by 97%), shortened the overall time dedicated to the evaluation (by 79%), and reduced the carbon footprint (by 93%). The majority of patients (94%) were highly satisfied with the remote evaluation service.

Kloosterman et al⁹ extended the concept of remote CIED management to the outpatient setting, specifically

Table. Summary of the Published Experiences With IRP Reporting Complication Rates

	Programmers	Software	Context	Patients	On site	Remote	Key findings
Kloosterman 2019	Medtronic 2090	RemoteControl Bomgar	MRI scan	50 (100 IRP)	MRI tech/ nurse	Cardiologist	No complication Patient satisfaction: 91%
Okabe 2020	Medtronic 2090	RemoteControl Bomgar	EP laboratory	4	EP staff	CIEDs tech	No complication
Siddamsetti 2021	Medtronic 2090	Bomgar	MRI scan	209 (418 IRP)	MRI tech	CIEDs tech	No complication
Xiong 2022	Abbott Merlin 3650	China Telecom Corporation	Emergency	3	Medical staff	Cardiologist CIEDs tech	No complication
Tong 2022	Abbott Merlin 3650	China Telecom Corporation	Hospital FU	325	Medical staff	Cardiologist	No complication Patient satisfaction: 90%
Kloosterman 2023	Medtronic 2090	RemoteControl Bomgar	At home FU	150 (300 IRP)	CIEDs tech Nurse Doctor	Cardiologist	26 device interrogation interruptions 1 interruption during the session No clinical adverse event Patient satisfaction: 85%
Alexander 2023	Medtronic 2090	RemoteControl Bomgar	ICD deactivation Hospital Home	6	Patient surrogate	Cardiologist	No complication
Ploux 2023	Abbot Merlin 2 Biotronik Renamic MicroPort Smart Touch	Teamviewer	Hospital FU MRI EP Lab	110 (115 IRP)	Cardiologist	Cardiologist	No clinical complication Patient satisfaction: 95%
Ploux 2024	Abbot Merlin 2 Biotronik Renamic Boston Scientific Latitude Medtronic CareLink SmartSync MicroPort Smart Touch	VPN	Hospital FU vs Remote FU	34	Nurse	Cardiologist	1 interruption during the session No clinical complication Patient satisfaction: 94% 97% reduction in travel expenses 93% reduction in carbon footprint 79% reduction in time spent

CIED indicates cardiac implantable electronic device; EP, electrophysiology; FU, follow-up; ICD, implantable cardioverter defibrillator; IRP, indirect remote programming; MRI, magnetic resonance imaging; and VPN, virtual private network.

for home-based interventions. Their protocol involved a technician visiting the patient's home to set up both a telehealth tablet for patient interviews and a dedicated Medtronic 2090 programmer equipped with the remote-control software. A total of 300 remote-control sessions were completed in 150 patients (99 pacemakers and 51 ICDs), of whom 81% considered the IRP to have superior advantages over the standard system of an in-office check. This study expands the scope of remote CIED management by demonstrating that home-based interventions are both feasible and highly convenient for patients, significantly reducing the need for travel and in-person hospital visits, an important benefit for elderly patients and those with mobility impairments.

In response to the COVID-19 pandemic, Okabe et al⁵ used IRP for device implantation and generator replacement procedures in 4 patients. The electrophysiology staff initiated device connectivity through the standard programmer interface, after which the remote device specialist performed comprehensive device testing functions. The technology provides immediate support from the device specialists, without exposing them to infectious risk or radiation exposure.

IRP was also used in the context of palliative care to deactivate the ICD therapies, allowing patients to remain

in their chosen place of dying at the end of life.²⁴ A proof of concept was done by the team of Alexander et al, where the local person was a patient surrogate.

ADVANTAGES

IRP requires the use of a programmer, which represents an important barrier against access to the CIED. The physical presence of a healthcare professional at the patient's bedside remains mandatory to initiate the connection between the device and the programmer. It is not possible to start a session without the patient's consent or the remote operator's secure identification. Any malicious access would need to occur during the interrogation session itself, under the immediate observation of the local personnel, who can terminate it at any point and can assume control of the programmer if necessary. This arrangement also ensures an immediate response should complications arise during device interrogation or subsequent reprogramming, for instance, loss of capture following pacing amplitude adjustments or oversensing after sensitivity modifications. IRP has been shown to be a safe alternative to in-person interrogation of CIEDs using different systems for different CIEDs in various clinical situations. Indeed, among a total of >1290 IRP

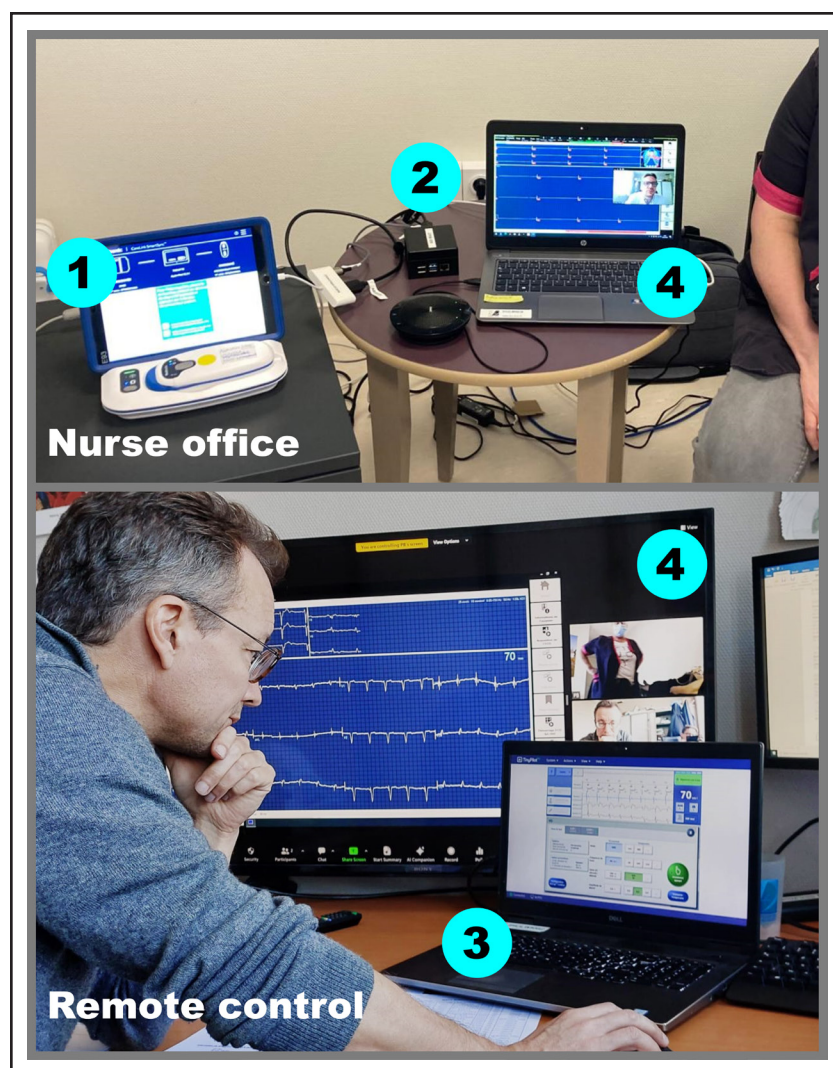


Figure 3. Setup of a custom multivendor indirect remote programming solution.

At the patient side, the cardiac implantable electronic devices (CIEDs) programmer (1) is connected to a minicomputer (2) using a VGA or HDMI for screen capturing and using a USB cable for cursor control. A secure connection was created using virtual private network service between the minicomputer and the remote control workstation (Wi-Fi or 4G) (3). A direct 2-way audio and video feed ensured communication between the remote control center and the patient using separate workstations (4) which were also used to display and register the 12-lead ECG. Reprinted from Ploux et al¹¹ with permission.

sessions, no clinical complications were reported. There were 28 communication interruptions (2%) due to an unstable internet connection. Although a strong internet connection appears to be critical, distance is not a concern. Kloosterman and our team have successfully completed several transatlantic IRP.

The preliminary studies conducted to date have strategically focused on clinical scenarios where RP addresses well-defined, recurring clinical challenges identified by medical teams, allied health professionals, and manufacturer's service representatives. IRP seems particularly relevant in outpatient settings, MRI suites, emergency departments, isolation wards, and radiotherapy sessions. In each of these environments, IRP provides a rapid, reliable alternative to on-site intervention, minimizing procedure delays. Recently, Biotronik LiveSupport was evaluated by Prasad et al in a survey assessing its utilization and user satisfaction. Between January 2022 and April 2025, the system was implemented across 47 centers in the United States. LiveSupport was used primarily for remote clinic follow-ups (54%), unplanned programming changes (13%), and remote MRI mode

programming (12%). During the study period, the number of monthly sessions increased progressively, exceeding 2000 sessions per month and reaching a total of 36369 sessions. This rapid adoption was attributed to a high level of satisfaction among users, including both healthcare providers and industry representatives. According to industry personnel, the use of LiveSupport saved more than 60 minutes per patient (including travel time) in 78% of sessions.²²

IRP conducted outside the hospital in a nurse office close to the patient's home or via a healthcare professional traveling to the patient's home seems safe and feasible and appears particularly well-suited to select clinical scenarios.⁹ For patients, especially the elderly or those with mobility limitations, IRP minimizes the burden of travel to specialized centers. This is especially impactful for individuals in rural or underserved regions, where access to electrophysiology expertise may be scarce. IRP ensures that these patients receive equivalent care without geographic barriers. Remote care can also improve access to specialized device management even in urban environments where mobility constraints can limit access

just as much as geography. The COVID-19 pandemic prompted exploratory evaluations of IRP as a critical tool for minimizing in-person contact while preserving continuity of care.^{5,8} This model enables uninterrupted CIED management and helps reduce staff exposure to viral transmission. Beyond the pandemic, this strategy shows promise for immunocompromised or high-risk individuals, for whom minimizing hospital visits remains a clinical priority. This approach is also highly adaptable to special populations, including incarcerated individuals or patients residing on remote islands. In such cases, the long-term placement of a programmer in a local nurse-run clinic may be far more efficient than requiring regular cardiologist travel to the region.²⁵

IRP also has the potential to redefine workflows, allowing device specialists and cardiologists to operate not only within hospitals but also remotely, from home, a model already established in radiology. This capability reduces the need for repeated on-site visits during nights or weekends by on-call staff, improving both staff efficiency and quality of life.

Last but not least, IRP was consistently associated with a high level of patient satisfaction, ranging from 85% to 95% in the different published experiences (Table).

LIMITATIONS

Given its encouraging safety profile and its potential clinical, logistical, ecological, and economic advantages, this model of IRP could reasonably expand its role in clinical practice over time. However, broad adoption will depend on resolving several challenges, including regulatory approval, cybersecurity standards, and integration into routine clinical workflows. Technically, a multivendor solution has been proposed and is already operational, though likely transitional. Ultimately, it is reasonable to expect each manufacturer to provide a secure, user-friendly platform for IRP of its own devices. The fact that the system requires a programmer makes it unsuitable for emergency situations, in particular when a problem is identified by RM. The first experiences with IRP highlighted the importance of a strong and stable internet connection, with a recommended speed of at least 10 Mbps.¹¹ CIEDs are particularly well-suited to remote evaluation, as they require minimal patient interaction and do not necessitate a physical examination. Nevertheless, for a limited number of patients, the preference for more human interaction was the primary reason for choosing in-person evaluation. Therefore, it is important to maintain real-time communication with the patient during the IRP session, at least by phone, but preferably using an audiovisual system. Some tests/functions of the device may pose a problem when performed remotely. During the FDA evaluation process of the Medtronic Remote Control system, the FDA suggested that the sponsor lock out functions that were particularly risky if used in

a remote-control situation (cardioversion, electrophysiology studies/arrhythmia inductions, underlying rhythm tests).²⁶ This is why users must agree not to perform these functions to obtain a session key from the software. On our side, we refrained from using the AAI mode during IRP and always required an external defibrillator to be present at the patient's bedside.^{10,11}

DIRECT RP

Technology

The term DRP refers to the process of programming a device remotely, without the need for an intermediary interface.

DRP of therapeutic CIEDs (pacemakers and defibrillators) has been implemented for a while, but for security reasons, it is strictly limited to specific functionalities. The communication pathway uses the same hardware as for remote interrogation, but in the opposite direction. At present, only the following tasks have been allowed to be managed remotely in some therapeutic CIEDs:

- Remote transmission scheduling
- Near real-time transmission of the device interrogation with a contemporary EGM
- Activation/deactivation of the patient-initiated transmission feature
- Activation/deactivation/resetting of alerts
- Reset of the episode logbook

Direct reprogramming of therapeutic CIEDs is therefore not a matter of technology but rather a security concern, which is discussed in a dedicated section.

Implantable cardiac monitors (ICMs), in contrast, have strictly limited diagnostic capabilities, alleviating the fear of harm in the event of communication failure or interception. The Medtronic Linq II ICM became the first device to allow DRP of alert settings, followed by the Boston Scientific Lux-DX ICM and the Abbott Assert-IQ ICM. This technology uses a patient transmitter (a smartphone with a proprietary application) to enable 2-way communication between the server and the patient's ICM. These devices can be fully reprogrammed remotely, allowing for comprehensive device management without requiring an in-office patient encounter.

EXPERIENCE WITH DRP

Due to the limited availability of DRP options for therapeutic devices, no study has been conducted to assess their usefulness. However, with the recent introduction of DRP functionality, real-world data on its use for ICM management have now become available. In 2 different cohorts involving Medtronic Linq II (19525 patients, mean follow-up of 8 months) and Boston Scientific Lux-Dx ILRs (8238 patients, mean follow-up of 182 days), it has been shown that 18.3% and 20% of the devices

were reprogrammed remotely, which represented 74.5% and 83.5% of the reprogramming events, respectively.^{12,13} In a smaller cohort involving 117 Medtronic Linq II and 105 Boston Scientific Lux-Dx, Neiman et al²⁷ observed a DRP rate of 15.4% and 35.2%, respectively, at 3 months. DRP resulted in a decrease from 0.13 to 0.03 median alerts/d and from 0.15 to 0.01 median alerts/d with the Linq II and the Lux-Dx, respectively. Emerging evidence supports the hypothesis that DRP of ICMs is a clinically valuable tool that can enhance device programming optimization while reducing the workload associated with patient follow-up. In fact, DRP represents one of the 2 key innovative strategies with artificial intelligence that are essential to ensure the long-term sustainability of ICM RM, given the substantial number of alerts generated, many of which are ultimately false positives. Notably, reprogramming is a Class I recommendation for ICMs on RM when frequent undersensing or oversensing occurs, to minimize future false-positive transmissions.¹

POTENTIAL ADVANTAGES OF DRP FOR THERAPEUTIC CIEDS

To date, all proposed RP solutions rely on a model in which a trained healthcare professional is physically present at the patient's side, while the actual device programming is controlled remotely by an off-site clinician. As previously discussed, IRP addresses some of the logistical barriers currently limiting postimplant CIED management. Although this model offers essential safeguards, its main limitation lies in the requirement for both a programmer and a trained healthcare professional to be physically present with the patient. This constraint significantly hinders its scalability and accessibility. Implementing large-scale IRP via home visits by healthcare professionals appears impractical.

The prospect of DRP, enabling device adjustments to be performed at a distance without the need for an on-site healthcare professional, represents an emerging paradigm in the management of CIEDs. This approach would offer substantial organizational and clinical benefits. It would reduce logistical burdens and human resource requirements while enabling timely interventions in critical situations, such as elevated capture thresholds in pacemaker-dependent patients, inappropriate ICD shocks, or inadvertent deactivation of device therapies following a procedure.²⁸ Device parameter adjustments could be implemented in near real time, potentially preventing symptomatic deterioration or the need for emergency hospital admissions. From this perspective, the current trend toward replacing a dedicated home transmitter with the patient's smartphone represents a particularly advantageous technological option, as it provides greater flexibility for device reprogramming outside the home environment.²⁹ Routine adjustments of

noncritical settings (eg, alert thresholds, rate-adaptive parameters) could be streamlined, reducing the strain on outpatient clinics. As seen with ICM, DRP could similarly optimize the management of programmable alerts, thereby reducing the burden on monitoring centers. Furthermore, this capability would facilitate the rapid deployment of essential software or firmware updates, particularly in the context of device recalls. The adoption of direct RP would represent a major advancement in postimplant care. Patients could benefit from continuous, optimized follow-up without the need to return to their implanting or managing center, thereby improving both healthcare efficiency and patient convenience. These benefits have already been demonstrated with the implementation of direct programming for neurological implantable devices.^{30,31}

POTENTIAL LIMITATIONS OF DRP FOR THERAPEUTIC CIEDS

The primary limitation of RP systems concerns patient safety, which may be compromised both by the risk of cyberattacks and by the possibility of procedural incidents occurring during remote interrogation or programming. Cybersecurity is a legitimate concern, particularly in the context of telecommunicated medical devices.^{28,32} No system can be considered entirely invulnerable; indeed, security vulnerabilities have been identified in both programmers and remote transmitters. However, to date, no cases of hacking involving a cardiac implantable device with potential adverse consequences for patients have been reported.³³ Technological advances in cybersecurity, along with improvements in the computational capabilities embedded within implantable devices, are expected to enable the development of highly secure and deterrent communication protocols. Regulatory frameworks and market authorization guidelines related to cybersecurity are evolving rapidly on both sides of the Atlantic.^{34–37} Nevertheless, it is likely that manufacturers seeking to develop RP functionalities will implement security measures that substantially exceed the minimum regulatory requirements.³⁸

Patient safety represents a critical concern when CIEDs are managed remotely in the home setting. Interrogation and programming of CIEDs can occasionally precipitate symptoms such as pacemaker syndrome, presyncope, or syncope, most commonly attributable to programming errors. Although exceedingly rare, the induction of ventricular arrhythmia during device interrogation has also been documented.³⁹ Although such complications occur infrequently, their consequences could be considerably more severe if they arose in an unattended patient at home. The central question is therefore whether RP would provide sufficient clinical benefits to outweigh this incremental risk at the population level.

One could argue that direct RP should be restricted to noncritical parameters (eg, refractory periods, pacing rate, sensors) to minimize the risk of adverse events. Conversely, clinically meaningful programming changes may be required to address significant clinical problems, such as adjustments to detection zones, counters, or therapies for ventricular arrhythmia. Contemporary CIEDs already perform several automated programming modifications autonomously, including assessments of capture and sensing thresholds, adjustments to pacing configurations in cardiac resynchronization therapy devices, atrioventricular delay and PVARP modifications, lead polarity switching, and antitachycardia pacing deactivation. Some devices have also implemented transient, automatically triggered programming changes during remote interrogation sessions. For instance, legacy Medtronic Adapta pacemakers delivered 17 seconds of asynchronous pacing (VOO or DOO mode) to generate a magnet ECG report during each remote interrogation. Similarly, Biotronik pacemakers released before the Sky platform periodically generated 2 sequences of 12 seconds of encouraged sensing and forced stimulation by applying temporary changes to the basic pacing rate and AV delays.⁴⁰ These autonomously operating algorithms have occasionally been associated with symptoms and, exceptionally, with serious adverse events; nevertheless, this level of risk has been considered acceptable by both industry and the cardiology community.^{41,42}

Defining programming priorities and potential restrictions is beyond the scope of the present work. However, it is reasonable to assume (at least initially) that new programming rules would apply to remote operations. Electrophysiological testing functions and arrhythmia induction capabilities would likely remain disabled for remote use. In pacemaker-dependent patients, additional safeguards may warrant consideration. For example, preventing programming to single-chamber atrial pacing modes, limiting reductions in ventricular pacing output amplitude, or restricting modifications of pacing polarity when the lead is uncoded or registered as unipolar. Another possible approach would be to allow certain programming changes only on a time-limited basis, such as the temporary deactivation of antitachycardia therapies.

In light of these potential risks and given the intrinsic nature of a medical procedure performed at a distance, the question of informed patient consent requires special attention. In conventional device interrogations, the patient's physical presence implies implicit consent to the procedure. However, when RP extends to therapeutic modifications, ethical and legal principles may necessitate explicit patient involvement in the initiation of the session following clear and appropriate information provided by their healthcare provider. This paradigm differs from current practice with implantable cardiac monitors, in which remote data transmission routinely occurs without active patient participation.

PERSPECTIVES

More than 2 decades after the introduction of communicating CIEDs, we may be entering a new era in which RP complements RM in the management of these devices. Following cardiologist-driven initiatives of IRP, which have demonstrated important organizational benefits and consistently high patient satisfaction, this strategy may become more widely adopted with the emergence of dedicated industrial solutions. The benefits already observed in terms of improved access to care, reduced interventional delays, decreased patient transportation, and a lower carbon footprint could be further amplified and extended to emergency situations with the development of DRP solutions. However, the transition to DRP cannot rely on technological readiness alone. Its implementation will require careful system design to address significant safety concerns, along with rigorous stepwise validation. The development of dedicated regulatory pathways for RP capabilities, providing clear guidance on liability, authentication protocols, and fail-safe mechanisms for remote interventions, will likely be essential to ensure clinical adoption. Collaboration between regulatory agencies, professional societies, and device manufacturers may therefore be necessary to establish harmonized standards to enable the safe deployment of RP technologies. The 2023 HRS/EHRA/APHRS/LAHR expert consensus statement on practical management of the remote device clinic endorsed this position, regarding experiences with IRP as “small forays into remote programming [that] need to be expanded to the point that they do not require the patient to be in a healthcare facility.”¹

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